

## CSCE 649 PHYSICALLY-BASED MODELING · MODULE 11 · LECTURE NOTES

# Fluids I: Grid-Based

*Navier–Stokes, split into pieces you can actually solve.*

## MODULE OUTCOMES

State the incompressible Navier–Stokes equations and interpret each term.

Explain operator splitting and why it makes the problem tractable.

Implement semi-Lagrangian advection and explain its dissipation.

Implement pressure projection and explain what it enforces.

Explain the MAC grid and why staggering matters.

## 1 The equations

$$\frac{\partial \mathbf{u}}{\partial t} = -(\mathbf{u} \cdot \nabla) \mathbf{u} - (1/\rho) \nabla p + \nu \nabla^2 \mathbf{u} + \mathbf{f}$$

$$\nabla \cdot \mathbf{u} = 0$$

| Term                                    | Meaning                                                                                                      |
|-----------------------------------------|--------------------------------------------------------------------------------------------------------------|
| $-(\mathbf{u} \cdot \nabla) \mathbf{u}$ | <b>Advection.</b> The fluid carries its own velocity along with it. Nonlinear, and the source of turbulence. |
| $-(1/\rho) \nabla p$                    | <b>Pressure gradient.</b> Fluid accelerates from high pressure toward low.                                   |
| $\nu \nabla^2 \mathbf{u}$               | <b>Viscosity.</b> Diffusion of momentum, smoothing velocity differences.                                     |
| $\mathbf{f}$                            | External forces — gravity, buoyancy, user forces.                                                            |
| $\nabla \cdot \mathbf{u} = 0$           | <b>Incompressibility.</b> No net flow into or out of any region, so the fluid's volume is preserved exactly. |

**PRESSURE IS A LAGRANGE MULTIPLIER, NOT A STATE VARIABLE**

**The second equation has no time derivative.** It is a constraint that must hold at every instant, not an evolution equation.

**Pressure is the force that enforces it** — precisely the role  $\lambda$  played in Module 04. Pressure is not a property of the fluid being carried along and evolved; it is *solved for*, from scratch, every step, as whatever field makes the velocity update respect incompressibility.

**This is why the pressure solve is a global linear system.** Incompressibility couples the entire domain instantaneously: squeeze the fluid in one place and it must move somewhere else *now*, however far away. There is no locality to exploit.

**And it is why the pressure projection is the dominant cost** of almost every fluid simulation. Everything else is local; this is not.

## 2 Operator splitting

Jos Stam's 1999 'Stable Fluids' is the most influential paper in graphics fluid simulation, and its contribution is a decomposition: rather than solving the full equation at once, apply each term in sequence, each with a method suited to it.

- 1 **Add forces.**  $\mathbf{u} += \Delta t \cdot \mathbf{f}$ . Trivial.
- 2 **Advect.** Move the velocity field along itself. **Unconditionally stable if done semi-Lagrangianly**, which is the paper's other contribution.
- 3 **Diffuse.** Apply viscosity by an implicit solve. **Often skipped entirely** in graphics, because the numerical dissipation introduced by advection already exceeds any physical viscosity you would have wanted.
- 4 **Project.** Subtract a pressure gradient so the field becomes divergence-free. **The expensive step.**

**Splitting introduces error** — applying the terms sequentially is not the same as solving them simultaneously, and the error is first order in  $\Delta t$ . It is accepted universally, because each individual step is tractable and the combined method is stable, which the unsplit equation is not.

**SEMI-LAGRANGIAN ADVECTION CANNOT BLOW UP**

The intuitive approach to advection pushes values forward: this parcel of fluid is moving, so move its velocity to where it will be. That is unstable at large time steps, because parcels can overtake one another and pile up.

**Stam traced backwards instead.** For each grid cell, ask where the fluid now in this cell came from one step ago — follow the velocity field backwards — and interpolate the old field at that position.

**The result is a weighted average of values that already existed**, so the new value cannot possibly exceed the range already present in the field. **It cannot blow up, at any time step.** This removed the CFL restriction that had made earlier methods expensive, and it is why the paper is titled 'Stable Fluids'.

**The cost is severe numerical dissipation.** Every step performs an interpolation, every interpolation is a smoothing operation, and the smoothing accumulates. Smoke visibly loses its fine structure within a few seconds, and the velocity field loses its small-scale vortices — which are exactly what makes smoke look like smoke.

| Technique                         | Mechanism                                                                                                   | Assessment                                                                                                                                                 |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Vorticity confinement</b>      | Detect where vorticity is concentrated and add a force pushing energy back into those vortices.             | <b>Entirely unphysical</b> — it adds energy at the scale dissipation removed it. Cheap, effective, and universally used. Nobody pretends it is principled. |
| <b>BFECC / MacCormack</b>         | Advect forward, then advect the result backward, and use the discrepancy to estimate and correct the error. | <b>Roughly twice the cost, and substantially less dissipation.</b> Principled, and the usual choice when the budget allows.                                |
| <b>Higher-order interpolation</b> | Cubic rather than trilinear interpolation in the advection step.                                            | Moderate improvement; introduces ringing near discontinuities, which must be limited.                                                                      |
| <b>FLIP</b>                       | Carry velocity on particles and avoid interpolating the field at all.                                       | <b>The real answer, and why FLIP displaced pure grid methods.</b> Module 12.                                                                               |

### 3 The MAC grid

#### STAGGER THE VELOCITY ONTO CELL FACES

The **obvious layout** stores everything — pressure and all three velocity components — at cell centres. Computing the divergence then requires a central difference spanning *two* cells in each direction.

**That stencil cannot see a checkerboard.** A pressure field alternating high-low-high-low between adjacent cells has, by the two-cell central difference, exactly zero gradient — so the solver cannot detect or correct it. The result is the classic **checkerboard instability**: a spurious oscillating pressure field that the method is blind to.

The **MAC (marker-and-cell) grid** stores pressure at cell centres and each velocity component on the cell *faces* perpendicular to it. The x-velocity lives on the left and right faces, the y-velocity on the top and bottom, and so on.

Now the divergence of a cell is a difference between its own adjacent faces — a compact stencil that sees everything, so the checkerboard mode is not in the null space and cannot arise. **And the pressure gradient naturally lands exactly where the velocity components are stored**, so no interpolation is needed in the projection. The layout makes two problems disappear at once, which is why it is universal.

$$\nabla^2 p = (\rho/\Delta t) \nabla \cdot \mathbf{u}^* \quad \mathbf{u} = \mathbf{u}^* - (\Delta t/\rho) \nabla p$$

Solve the Poisson equation for the pressure whose gradient exactly cancels the divergence of the intermediate velocity  $\mathbf{u}^*$ , then subtract it. The matrix is sparse, symmetric, and positive definite, so conjugate gradient applies — and **this solve is where the time goes**. An incomplete Cholesky preconditioner is the standard choice; **multigrid** is asymptotically optimal and is what large production solvers use.

### 4 Liquids

|                        | Smoke                                            | Water                                                                                                                         |
|------------------------|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Domain                 | Fluid fills the whole grid. Every cell is fluid. | <b>A free surface</b> separating liquid from air, which moves and must be tracked.                                            |
| Boundary conditions    | Simple — the domain walls.                       | The free surface is a pressure boundary condition that changes every step.                                                    |
| Representation         | A density field advected with the flow.          | <b>A level set</b> — the surface is the zero set of a signed distance field.                                                  |
| Topology change        | Not applicable.                                  | <b>Automatic.</b> Splitting droplets and merging pools require no special handling, which is the level set's great advantage. |
| Characteristic failure | <b>Dissipation</b> — detail is lost.             | <b>Volume loss</b> — see below.                                                                                               |

**VOLUME LOSS IS THE PROBLEM TO MEASURE**

A grid-based liquid simulation loses volume steadily, from several compounding causes: the level set is advected by the same dissipative scheme as everything else; reinitialising it to a true distance field each step smooths it further; and any feature thinner than a grid cell simply cannot be represented and disappears.

**The visible result is that a splash's droplets vanish in mid-air** and a thin sheet of water evaporates. Over a long simulation a pool can measurably drain.

**Plot total volume every frame.** It is the single most informative diagnostic for a liquid solver — the exact counterpart of the energy plot for a dynamics solver — and it is what Project 2 asks for.

**Particle level sets** mitigate it by seeding marker particles near the surface and using them to correct the implicit representation where it has drifted. **And FLIP (Module 12) loses far less volume** by carrying the fluid on particles in the first place, which is the principal reason it displaced pure grid methods for liquid.

## Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

| Resource                                                    | What it covers for this module                                                                                                        |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Bridson — Fluid Simulation for Computer Graphics            | The standard reference for this module and the next. The free course notes cover the MAC grid, projection, and level sets completely. |
| Stam — Stable Fluids (1999, free)                           | The splitting and semi-Lagrangian advection of §2. Short, and it changed the field.                                                   |
| Stam — Real-Time Fluid Dynamics for Games (free, with code) | A complete 2D solver in about a hundred lines. Implement this first.                                                                  |
| Fedkiw, Stam & Jensen — Visual Simulation of Smoke (free)   | Vorticity confinement, from the paper that introduced it to graphics.                                                                 |
| Ten Minute Physics — Eulerian fluid episodes                | A working grid solver built up step by step, with code.                                                                               |

## Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Implement Stam's 2D solver from the 'Real-Time Fluid Dynamics for Games' paper. Get smoke moving before attempting anything else.

- 2 Implement a collocated grid and demonstrate the checkerboard instability. Then switch to a MAC grid and show it disappears.
- 3 Implement semi-Lagrangian advection and measure the dissipation: advect a sharp density blob in a uniform velocity field and plot its maximum value over time. It should decay — report how fast.
- 4 Implement BFECC or MacCormack advection and repeat. Report the improvement and the added cost.
- 5 Implement vorticity confinement and show the recovered detail. Then measure the energy it adds and be honest about it.
- 6 Implement pressure projection with conjugate gradient. Plot residual against iteration count and report what tolerance is visually sufficient.
- 7 Add an incomplete Cholesky preconditioner and report the reduction in iterations.
- 8 Measure what fraction of total simulation time the pressure solve consumes, at three grid resolutions.
- 9 Implement a level set free surface and simulate a dam break. **Plot total volume every frame** and report the loss over ten seconds.
- 10 Halve the grid spacing and repeat the volume measurement. Report how volume loss scales with resolution.

## Self-check

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Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 State the incompressible Navier–Stokes equations and interpret each term.
- 2 Why is pressure a Lagrange multiplier, and what does that imply about the solve?
- 3 List the four steps of operator splitting and say which is expensive.
- 4 Why can semi-Lagrangian advection not blow up, and what does it cost?
- 5 Name three ways of fighting advection dissipation and assess each.
- 6 What is the checkerboard instability and how does the MAC grid prevent it?
- 7 Write the pressure Poisson equation and say what is solved and what is subtracted.
- 8 Give three differences between simulating smoke and simulating water.
- 9 Why do grid-based liquids lose volume, and what should you plot?